

AWS COURSE CONTENT

Duration- 6-8 Weeks

WEEK - 01

Introduction to Cloud & AWS Fundamentals

- Introduction to Cloud Computing (IaaS, PaaS, SaaS)
- Overview of AWS Global Infrastructure (Regions, AZs, Edge Locations)
- Introduction to AWS Console & IAM
- Identity and Access Management (IAM): Users, Groups, Roles, Policies
- Hands-on: Creating a free-tier AWS account & IAM user setup

WEEK - 02

Compute Services

- Introduction to EC2 (Elastic Compute Cloud)
- EC2 instance types, AMIs, key pairs, security groups
- Launching and connecting to EC2 (Linux/Windows)
- Elastic Load Balancer (ELB) and Auto Scaling
- Practice: Hosting a website on EC2

WEEK - 03

Storage and Database Services

- Amazon S3 (Simple Storage Service): Buckets, Objects, Permissions
- Lifecycle policies, Versioning, Static website hosting
- Amazon EBS (Elastic Block Store) vs EFS
- Introduction to Amazon RDS & DynamoDB
- Practice: Backup and restore using S3

WEEK - 04

Networking and Content Delivery

- Virtual Private Cloud (VPC): Subnets, Route Tables, Internet Gateway, NAT
- Security Groups vs Network ACLs
- AWS Route 53: Domain Registration & DNS
- CloudFront (CDN) basics
- Practice: Configure a custom domain for a web app

WEEK - 05

Monitoring, Logging, and Automation

- AWS CloudWatch: Metrics, Alarms, Logs
- AWS CloudTrail for auditing
- Simple Notification Service (SNS)
- AWS Lambda (Intro to serverless)
- Practice: Trigger Lambda with S3 event

WEEK - 06

DevOps and Deployment Tools

- Infrastructure as Code: Introduction to AWS CloudFormation
- AWS CodePipeline, CodeBuild, and CodeDeploy
- Elastic Beanstalk for deploying applications
- Practice: Automate deployment of a sample app

OPTIONAL ADVANCED WEEKS

WEEK - 07

Security, Identity, and Compliance

- AWS Shared Responsibility Model
- AWS KMS (Key Management Service)
- AWS WAF and Shield (web application firewall, DDoS protection)
- Practice: Enable encryption and MFA for AWS services

WEEK - 08

Real-world Projects & Certification Prep

- Capstone Project: Deploy a full-stack web application using AWS services
- AWS Well-Architected Framework
- Tips for AWS certification exams [Cloud Practitioner/SAA-C03]
- Mock test and feedback session

Tools & Services Covered

- **Core:** EC2, S3, IAM, VPC, RDS, CloudWatch, Lambda
- **Developer Tools:** CodePipeline, CodeBuild, CodeDeploy
- **Security & Monitoring:** KMS, CloudTrail, Shield, WAF
- **Others:** Route 53, CloudFront, EFS, Elastic Beanstalk

Course Deliverables

- Slides & Lecture Notes
- Lab Guides & AWS Hands-on Tasks
- Practice Quizzes & Mock Exam
- Final Project Evaluation
- Certification Readiness Checklist